

Project Title: LUCID: Low-dose Understanding, Cellular Insights, and Molecular Discoveries

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Project Summary: High doses of ionizing radiation (above 50-100 mGy) are established carcinogens. However, at doses of less than 10 mGy, the risks of carcinogenesis are difficult to ascertain due to uncertainties arising from limitations in our understanding of the molecular and cellular mechanisms involved, lack of high-quality datasets connecting exposure to outcomes, and challenges in modeling the multiscale processes that connect radiation damage to tissue level responses. Low-dose (LD) exposures are common and unavoidable, arising from natural and medical sources, but their relationship to cancer risk remains an open question in radiobiology and public health. LUCID proposes *an integrated experimental, computational, and data-driven approach that harnesses the power of AI to create a comprehensive, integrative system capable of significantly automating hypothesis generation, paving the way for more efficient identification and prioritization of research questions. By integrating a diverse array of data and resources and applying natural language processing and advanced AI algorithms, our approach aims to uncover novel insights, suggest potential collaborations, and propose follow-up experiments or analyses—leading to breakthroughs in our understanding of this complex field.*

LUCID Responses to Reviewer Feedback

1) Reviewers were skeptical that enough data for low dose radiation research existed on which to build AI/ML framework at the scale and scope intended for this project. There were also concerns about the quality of data available and how that data is curated for inclusion in the AI/ML analyses.

We will be identifying and cataloging the available sources of low dose radiation and health data during of the first 6 months. During this time, the T1 team will be focused data collection, curation, assessment, and storage. We will perform a comprehensive search for publicly available repositories, including data from individuals with occupational exposure, medical radiation exposure, and environmental radiation exposure. We will also collect data from experiments using animal models, as it may be feasible to use it in our multimodal models.

All data will be processed following standard procedures and stored in a centralized location. We will consider the use of Radiation Biology Ontology (RBO) and other ontologies to ensure the consistency, data quality, and knowledge-based data analyses. We will iteratively refine our study design and research direction for the subsequent thrusts to realistically match our any data constraints that may exist. While there are certainly limitations with this approach, we believe that it is important to move forward with the data we do have as prospective studies are costly and time intensive.

Tumor tissue samples from longitudinal cohorts already number in the hundreds, this will be enough samples to identify gene expression signatures (<https://www.liebertpub.com/doi/10.1089/cmb.2012.0283>).

2) Some reviewers indicated that a wealth of previous low dose radiation research results (and sample archives such as human tissue archives) do not appear to be known to this group or at least not referenced. Lack of mention of the recent NASEM strategic plan recommendations

amplified this concern.

The team is aware of the scope of previous low dose research has generated, from the roughly 180,000 papers related in some way to low dose radiation research (including the 45,000 open access papers that we have assembled on low dose research), to the tissue archives. Since the vast majority of the archival samples have not been molecularly characterized, they pose a unique challenge to incorporate into our models. Processing existing samples is not in scope of the current proposal, though it is something that could be considered. Techniques have advanced recently to enable reliable molecular readouts on preserved samples.

Regarding the NASEM report and recommendations. The team is well aware of them and this proposal addresses a number of the recommendations, particularly the recommendations to leverage the modeling and simulation and machine learning to advance data integration for low dose research.

3) While there is considerable expertise to carry out the computational aspects of the proposal there appeared to be little low dose radiation biology expertise on the project.

Low-dose radiation biology expertise is available within our larger group through collaborations with academic partners. Now that the project is started, we will activate our network of collaborators, which includes radiation biology expertise at multiple partner locations. For example, ORNL has long standing collaboration with radiation biologists at Duke University (Radiology and Radiation Oncology) and UTK (Cell and Mol Bio) on guiding the development of computational models of cells based on biological information. We expect to continue to build and leverage these collaborations while performing the work proposed on the LUCID project.

4) The experimental aspects of the proposal lacked sufficient detail to justify the choices of cell types and the relevance of those choices to low dose research. Much work has already been done in single cell exposures, less so in organoid systems. A clearer explanation of what new science insights would be gained relevant to low dose research was needed to justify the work.

LUCID's experimental focus is to collect data that supports hypothesis formation and testing on molecular mechanisms of low dose, including data that support our thrusts in identification of low dose biomarkers and mutational signatures. We expect these experimental studies will inform low dose impact on cellular function, but they will also offer a robust integration of imaging and multi-omic data and support efforts to validate biomarkers. This data will also be utilized in training and testing AI-driven and mechanistic models of disease processes.

On the selection of cell model systems, we've chosen Human Umbilical Vein Endothelial Cells (HUVECs) for their unique relevance to low-dose radiation biology, given they are primary cells that lack the genetic modifications resulting from the cell immortalization process. This allows a more accurate representation of human genetic variations in radiation response experiments while establishing high-throughput cell culturing and assay techniques. Additionally, our focus extends to organoids, which provide a more complex yet invaluable representation of cellular interactions, unattainable with 2D cell model systems. Our use of human induced pluripotent stem cells (iPSCs) will provide insights into organoid radiation responses, offering a systems biology understanding facilitated by AI-guided models.

5) While themes 1, 3, 5 and 6 seem achievable in the timeframe suggested, themes 2, 4 and 7 appeared to be longer term goals. In particular, themes 4 and 7 appeared to be longer term goals than this 3-year effort could support. Consider scoping the project back to emphasize themes 1, 3, 5 and 6.

In response to this guidance, we are taking the following steps.

Thrust 2 is refocused on extracting and organizing low dose hypotheses from the literature (rather than the other sources) and providing a robust AI workflow feeding structured data into theme 3. Theme 2 has already curated 45K papers that will be used to feed this workflow. This work can clearly be completed in the three years.

Thrust 4 is refocused on developing software for estimating “value of (new) information” (can be new simulation output or new wet-lab experimental data). The “Value of information” (Vol) metric is defined in terms of the expected ability of the information source to constrain a predictive model (in terms of reduced uncertainty in its parameters or other inputs). Vol will be computed by comparing ensembles of model simulations against simulated data. We will use the multiscale modeling framework developed in Task 7 as an exemplar. Milestones have been adjusted to reflect that reduction in scope.

Milestones:

- Ensemble studies using Task 7 model(s) to quantify sensitivity of observables to model inputs
- “Simulated experiments”: add noise to model predictions to simulate experimental measurements
- Value of information: study how constrainable the model becomes as different information sources or signal-to-noise ratios are varied
- Software to compute Vol in generalized settings

Thrust 7 focuses on developing a digital-twin-based modeling and simulation platform to predict low dose radiation effects in populations of humans exposed to different dose scenarios. The work builds on considerable progress made over the last 3-5 years on establishing a radiation dosimetry simulation pipeline along with virtual human phantoms to assess damage from low dose radiation. The current phase of the work will enhance the simulations by adding biochemical processes to the simulation platform. We recognize this is an ambitious task and are treating it appropriately. In light of the reviewer’s comments, we will reduce the scope of T7 by removing the synthetic data generation task and relegating it to a future effort. T7 will now focus largely on the continued development of the simulation platform and extensive validation through experiments performed in T5.

6) The PIER plan lays out approaches for outreach and recruitment of underrepresented groups but does not fully address how integral these efforts will be across all partners.

Under the coordination and oversight of the LUCID PI, Partnerships and Outreach Manager, Steering Committee, and Project Manager, LUCID will ensure tightly coupled and synergistic DEIA activities across both all seven thrusts of the project and all project partners to drive LUCID’s proactive approach to training, recruitment, hiring, and retention. As part of the first 90 days of the project each thrust will develop a plan for PIER, these plans will be shared, and integrated into a common approach for the whole project. We will monitor progress on our PIER plan in our quarterly project wide meetings and adjust the plan based on progress.